

SpinExplorer: A python based application for converting, processing, viewing and analysing 1D,2D and 3D NMR spectra

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SpinExplorer is a Python-based graphical interface that allows raw Bruker/Varian NMR data to be converted to viewable NMR spectra. The data conversion and processing routines can be performed using `nmrPipe`¹ (if installed) or `nmrglue`². Processed spectra can then be viewed, overlaid, and analysed. 1D/2D Spectra processed using TopSpin can also be viewed and analysed.

The SpinExplorer package consists of three sub-packages called SpinConverter, SpinProcess and SpinView. SpinConverter provides a way to convert raw NMR data to `nmrPipe` format. It is similar to the Bruker/Varian commands on `nmrPipe` and allows a user to check spectral parameters before converting the spectra. (Note this feature is in development and needs updating to convert Bruker data formats from newer spectrometers)

SpinProcess allows the selection of desired processing parameters such as digital solvent suppression, apodization (window functions) and phasing. The data is then processed to produce an NMR spectrum.

SpinView allows users to view 1D, pseudo2D, 2D, pseudo3D and 3D NMR spectra. It also has analysis packages for pseudo2D relaxation, diffusion and CEST data (with the hope of adding more in the future).

Test data (1D-3D, TestData.zip) is available to download from:

<https://github.com/bindresearch/SpinExplorer>

Video tutorials for the program can be viewed at <https://www.youtube.com/@BindResearch>.

¹F. Delaglio, S. Grzesiek, G. W. Vuister, G. Zhu, J. Pfeifer and A. Bax: NMRPipe: a multidimensional spectral processing system based on UNIX pipes, J. Biomol. NMR. 6, 277-293 (1995).

²J.J. Helmus, C.P. Jaroniec, NmrGlue: An open source Python package for the analysis of multidimensional NMR data, J. Biomol. NMR 2013, 55, 355-367, <http://dx.doi.org/10.1007/s10858-013-9718-x>.

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0.1 Installation from released versions (recommended for most users)

On the SpinExplorer GitHub releases page (<https://github.com/bindresearch/SpinExplorer/releases>) download the .zip file associated with your operating system/chip architecture. Downloadable applications are currently available for MacOS (with Apple silicon or Intel-based architectures) and Windows (Intel/AMD-based architectures).

Once downloaded, extract the zipped application and move the entire folder to the applications folder. Double click on the application entry point (e.g. SpinExplorer_MacOS_AppleSilicon/SpinExplorer) to open the application. Note that your operating system may block the application from opening at this point. If this happens, do not move the package to the bin. Instead, go into Privacy and Security settings to allow the package to open. e.g. On MacOS, this block can be overridden by going into Privacy and Security, scrolling down to Security and clicking on "allow anyway".

Once allowed, the application will open to the home-screen (see section 0.3.

0.2 Installation from GitHub (for advanced users)

0.2.1 General

SpinExplorer is a Python-based app that can be used on Mac, Windows and Linux. The app requires python3 to be downloaded for a GitHub installation (3.11 or later). Clone the repository from GitHub at <https://github.com/bindresearch/SpinExplorer>.

0.2.2 MacOS/Linux

Making a virtual environment

It is advisory to make a virtual environment to place the application in order to prevent version issues with other applications or the system

Details at <https://docs.python.org/3/tutorial/venv.html>

To create a virtual environment, use the commands below in a terminal:

```
python3 -m venv SpinEnv
```

This virtual environment can then be activated using:

```
source SpinEnv/bin/activate
```

Anaconda environments can also be used instead if preferred.

Ensure the current working directory is in the main SpinExplorer directory (containing the project.pytoml) and then install SpinExplorer and its dependencies using:

```
python3 -m pip install .
```

The virtual environment needs to be activated every time a new terminal is created. The activation can be performed automatically by adding the line `source SpinEnv/bin/activate` to your `.zshrc` or `.bashrc` file.

MacOS/Linux installation

wxPython

It is worth noting that for Linux users wxPython does not have an official wheel for all versions on Linux. For x86_64 architectures, official wheels have been built for wxPython 4.2.3 on Ubuntu (20.04-LTS, 22.04-LTS, 24.04-LTS). Wheels for other Linux types such as Debian ($\geq 8, \leq 11$), Fedora ($\geq 23, \leq 38$), Rocky (8,9) and CentOS (7,8) are available for older wxPython versions ($< 4.2.3$).

More details at <https://wxpython.org/pages/downloads/>
and

<https://wxpython.org/blog/2017-08-17-builds-for-linux-with-pip/index.html>

0.2.3 Windows

The installation for Windows is similar to that for MacOS/Linux. Using py (py-launcher) is recommended for windows systems. If py is not present, download the latest python from <https://www.python.org>, and ensure install py-launcher is ticked. If desired, a virtual environment for the package can be created according to:

<https://docs.python.org/3/tutorial/venv.html>.

Install the SpinExplorer package and all dependencies using the following command in command prompt/power shell when in the cloned GitHub SpinExplorer directory:

```
py -m pip install .
```

0.3 Opening SpinExplorer

After clicking the SpinExplorer application (the application executable inside the downloaded application folder), the homepage will open. This enables users to view their recently opened data and open a file browser to search for experimental data to process or analyse. When the desired dataset has been selected, the SpinConverter, SpinProcess and SpinView programs can be opened in order to perform data processing and analysis. An introduction to each of these components of SpinExplorer are discussed below.



Figure 1: The main window of SpinExplorer

0.4 Converting Spectra

SpinConverter can be used to convert raw Bruker/Varian NMR FID data to nmrPipe format. It works in an analogous way to the Bruker/Varian commands in nmrPipe. SpinConverter can be accessed from the SpinExplorer interface, or it can be initialised in the terminal (MacOS/Linux) or Windows PowerShell/Command-Prompt (Windows) by running the command “SpinConverter” from a directory containing the raw NMR data. An example is shown in figure 2 for 1D proton data on a Bruker spectrometer. The code will read the relevant parameter file for Bruker (acqus) or Varian (procpa) and populate the SpinConverter window with the acquisition parameters used for the experiment. The SpinConverter window also allows the user to change these settings manually if they have not been accurately determined by the code.

In addition, the carrier frequency can be calculated using different referencing modes (H₂O, O1/BF₁ etc.). In multi-dimensional experiments (such as the [¹³N,¹H-HSQC]), the indirect dimension can be referenced from the water chemical shift. Additionally, if a multi-dimensional spectrum has been obtained using non-uniform sampling (NUS), the data can be re-shuffled to the correct order by selecting “Include NUS reconstruction”. Please note that this feature is only available on machines containing nmrPipe.

The set of conversion parameters can be saved (into parameters.json in the directory of the current dataset) and conversion can be performed using either nmrPipe (if installed) or nmrglue.

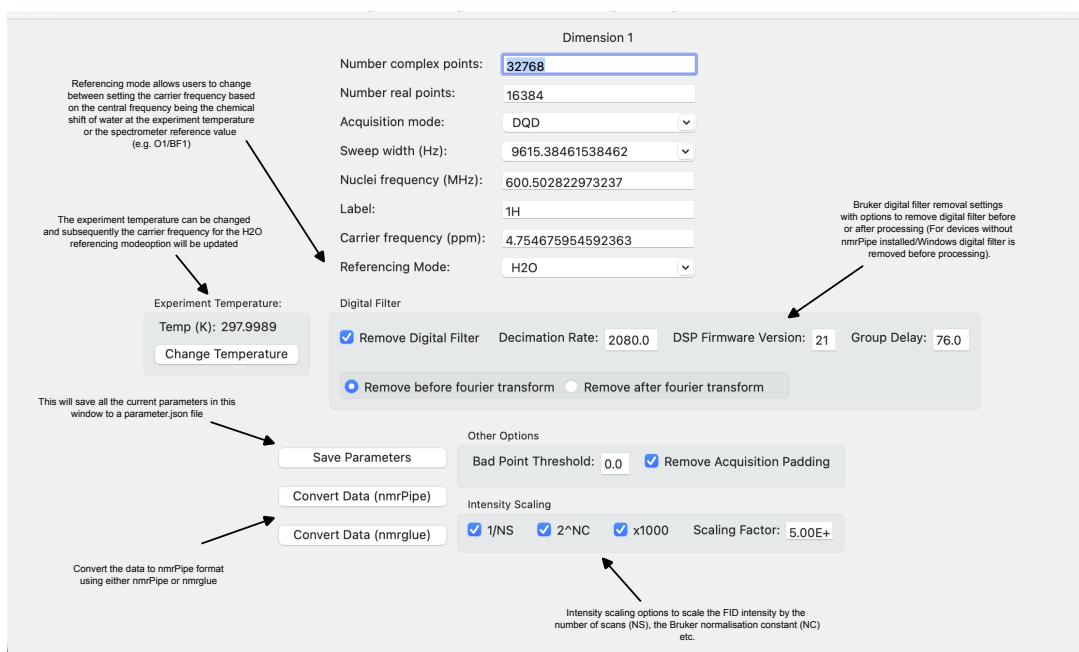


Figure 2: The window produced when SpinConverter is ran from a directory containing 1D Bruker Proton NMR data. The window is annotated with descriptions of a range of the features available.

0.5 Processing Spectra

Once the raw NMR data has been converted to nmrPipe format using SpinConverter, the data can be processed using SpinProcess. This can either be accessed from the SpinExplorer homepage or by running the command “SpinProcess” in the relevant directory in the terminal (MacOS/Linux) or Windows PowerShell/Command-Prompt (Windows). The window contains a tab for each dimension of the spectrum (excluding any pseudo axes containing different gradient strengths/time delays etc). Figure 3 shows the tab for the direct dimension of a Bruker 2D [¹³C, ¹H-HSQC].

Check-boxes are used to select for desired processing routines to be included when processing the spectrum. The information buttons provide pop-out windows once clicked containing information on each processing routine, along with links to further information. A wide variety of apodization/window functions can be selected whose parameters can be altered by the user. In addition, a plot of the first slice of the FID is displayed with an overlay of the selected window function. Clicking the plot will open it in a pop-up window. Furthermore, an interactive phasing window is available to determine the phasing parameters for the direct dimension of the NMR spectra. Finally, chemical shift extraction and baseline correction can be added to improve spectral quality.

For the indirect dimension, SMILE NUS reconstruction can be applied to the spectrum for non-uniformly sampled data. However, the reconstruction will only be performed accurately if the correct phasing parameters are inserted. For NUS data

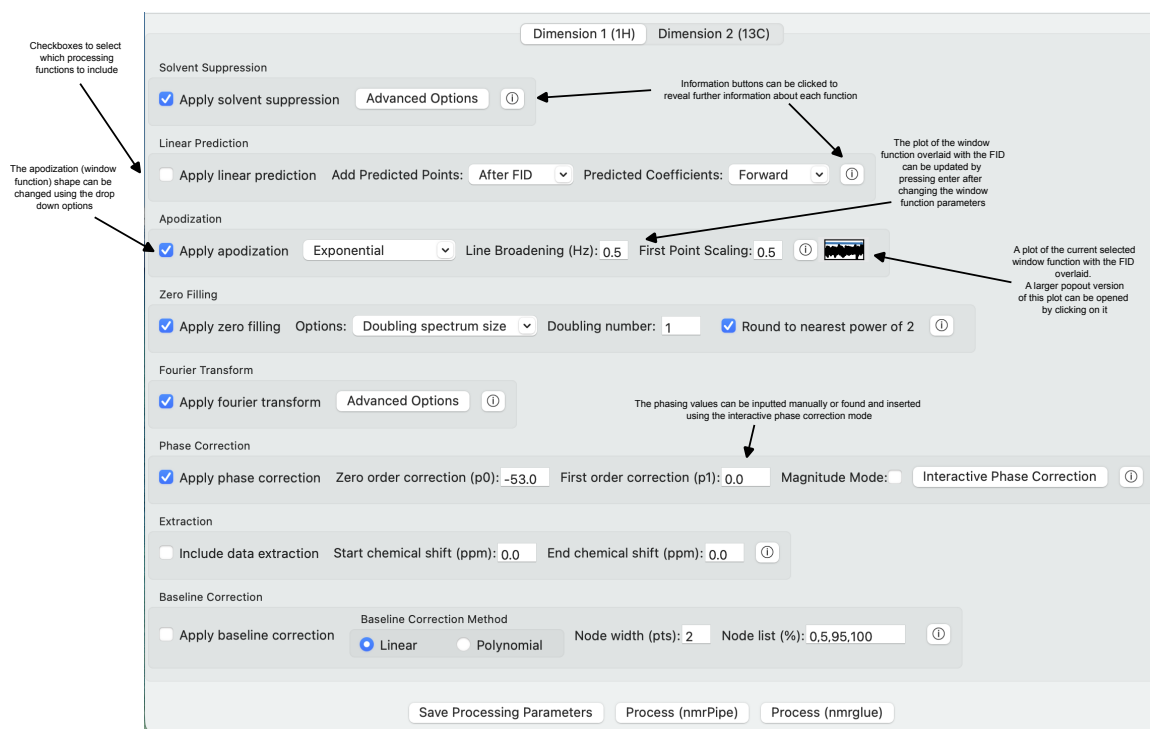


Figure 3: The first tab of a window produced when SpinProcess is ran from a directory containing a Bruker 2D [$^{13}\text{C},^1\text{H}$ -HSQC] which was converted to nmrPipe format using SpinConverter.

that has been successfully re-shuffled in SpinConverter, it is advised to first process the data without SMILE NUS reconstruction to find the correct indirect phasing parameters before re-processing with SMILE.

Once the desired parameters have been selected for each dimension, the data can be processed using the Process (nmrPipe) or Process (nmrglue) buttons. The parameters used for the processing can be saved, and will be located in a parameters.json file (along with any conversion parameters used) in the folder of the selected dataset.

Note that on machines without nmrPipe installed, non-uniformly sampled spectra cannot be processed in the current version of SpinProcess.

0.6 Viewing/Analysing Spectra

Once spectra have been converted and processed using SpinConverter/SpinProcess, NMR spectra can be viewed using SpinView. SpinView allows 1D, 2D and 3D NMR spectra to be inspected, re-processed, overlaid and analysed. For 2D/3D datasets, peak picking can be applied. For pseudo2D spectra, there are also frames for analysing diffusion/relaxation data. SpinView can be accessed from the Spin-Explorer main frame, or using the command "SpinView" from the terminal (MacOS/Linux) or Windows PowerShell/Command-Prompt (Windows) when in the di-

rectory containing the processed NMR data.

0.6.1 1D

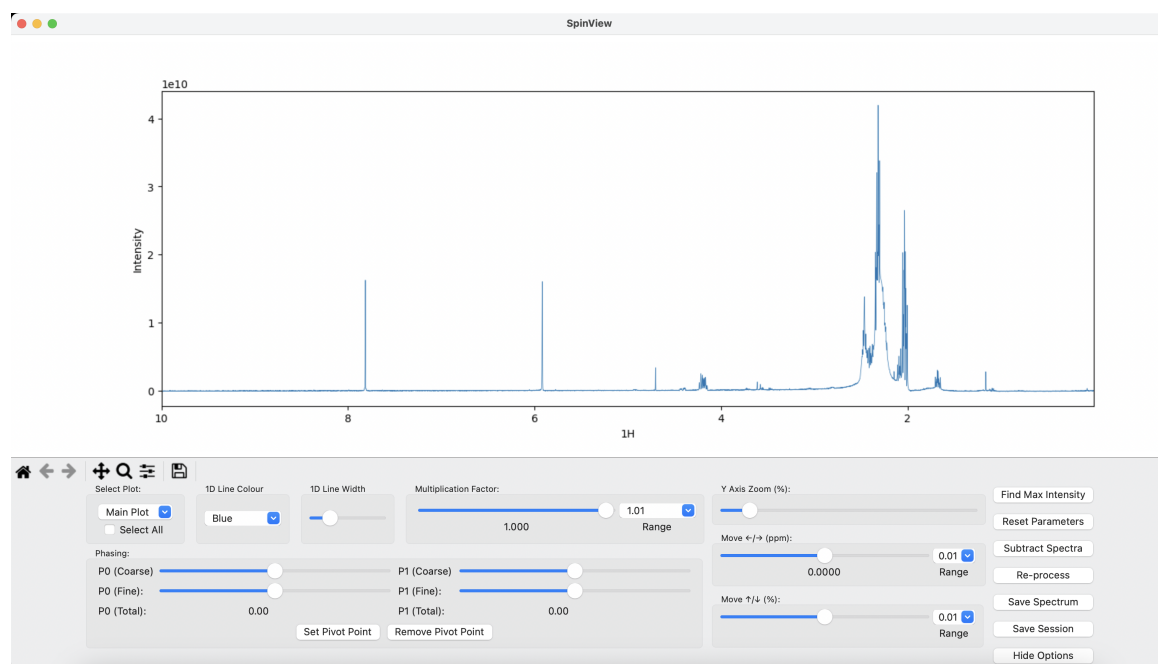


Figure 4: The SpinView window obtained when opened from a directory containing a processed 1D proton dataset

Figure 4 shows the window produced when SpinView is run from a directory containing a processed 1D proton spectrum. TopSpin processed data can also be opened running SpinView from the processed data folder (e.g. /pdata/1). The figure is a matplotlib-based figure with the standard matplotlib toolbar to zoom, scroll and save. It is possible for the user to re-phase the spectrum (using the coarse and fine controls) with the option to add a pivot point to determine an accurate first-order phase correction. Spectra can be moved left/right or up/down for alignment/baselining purposes. Additionally, spectra can be multiplied by a factor using the multiplication slider. The ranges of the move left/right, up/down, and multiplication sliders can be altered using the relevant drop-down boxes. It is possible to save the state of a manually edited spectrum using the “Save Spectrum” button. The original spectrum can be re-processed using the “Re-Process” button, which will open up an instance of SpinProcess. All manually changed parameters can be reset to their original values using the “Reset Parameters” button. For data that is difficult to baseline using the SpinProcess baselining options, manual spline baselining can be accessed using the “Baseline” button. The Y Axis Zoom slider can be used to change the scale of the intensity axis. Peak intensities and integrals can be obtained using the “Calculate Intensity” button.

Multiple spectra can be overlaid by dragging additional spectrum (.ft) files from a file browser onto the figure canvas. TopSpin processed data can also be added to

the canvas by dragging and dropping the processed data folder (e.g. /pdata/1) onto the canvas. Independent processing can be applied to each added spectrum, or the "Select All" checkbox can be clicked to apply processing to all overlaid spectra. The "Subtract Spectra" button allows different overlaid spectra to be subtracted from one another. The subtracted spectrum can then be saved as a separate spectrum. The "Save Session" button allows the current state of SpinView to be saved (as a .session file). When SpinView is run from a directory containing a .session file, the application will ask the user if they wish to open the saved session.

0.6.2 2D



Figure 5: The SpinView window obtained when opened from a directory containing a processed 2D [^{15}N , ^1H -HSQC] spectrum.

Figure 5 shows the window produced when SpinView is run from a directory containing a processed 2D [^{15}N , ^1H -HSQC] spectrum. Horizontal/vertical slices of the 2D spectrum can be accessed by clicking on the spectrum and pressing the "h" or "v" keys, respectively. Further clicking on the spectrum will enable different horizontal or vertical slices to be viewed. These slices can be phased like in the 1D frame. In addition, the "Y-Axis Zoom" slider can be used to zoom in on the currently selected slice.

The "Contour Start" slider can be used to update the intensity at which the 2D contour levels start. The number of contour levels can be changed using the "Contour Levels" slider. Furthermore, 2D spectra can be transposed using the "Transpose" button to view 2D spectra from an alternative perspective. The "Reset" button can be used to reset the manually altered parameters.

It is also possible to overlay 2D spectra in the same manner as 1D spectra. The "Move 2D Plot" sliders can move each overlaid spectrum independently, which can be useful for reference purposes. The "Save Session" button can be used to save the state of the SpinView instance, along with any overlaid spectra.

Finally, the "Stack Slices", "Fit Diffusion", and "Fit Relaxation" buttons are designed for pseudo2D spectra to provide further downstream analysis.

Peak Picking

The ability to read and pick peaks can be accessed from the "Read Peaks" button. Two-dimensional peak lists in tabular format obtained from external packages can be loaded, interactively adjusted, and then saved. Peak picking can be applied to the 2D plot using the nmrglue peak picking algorithms. The threshold for peak detection, along with the peak picking algorithm can be selected. Details of the peak picking algorithms can be found in nmrglue documentation. The picked peaks can again be interactively adjusted if the algorithm did not pick the desired peaks.

Stacking Slices 2D

Stacking 2D slices will produce a 1D SpinView window with each slice along the Y axis of the 2D plot overlaid. This window has the full functionality of the 1D SpinView window. Additional 1D spectra can be overlaid above these stacked slices by dragging and dropping the spectrum (.ft) files onto the figure canvas. The intensity/integral of a user-selected section of each slice can be determined using the "Find Max Intensity" button.

Fitting Diffusion Data

The diffusion fit window can be used to obtain the diffusion coefficients of peaks present in a diffusion NMR spectrum (Figure 6). Upon opening the window, the Bruker/Varian checkboxes can be used to change the spectrometer type. This enables the code to find the relevant parameters from the acquis/procar files when the "Find Parameters" button is pressed. If the bipolar gradients are selected, the δ value will be doubled automatically. The diffusion experiment delay parameters can also be added manually by the user.

The user can then input the value for the gradient integral factor (lower than 1.0 for shaped pulses) and the max gradient of the spectrometer used for the experiment. For Varian spectrometers, the DAC to Gauss/cm conversion parameter is required (set to 0.002 as default). The "Find Gradient Percentages" button will then attempt to automatically find the effective gradients used in the system from the spectrometer parameter files and the user-inserted values. Alternatively, these values can be inserted manually using the "Input Manually" button.

A user can then press the "Select Noise" button before dragging over an area of the spectrum containing no NMR peaks. The selected noise region will be shown

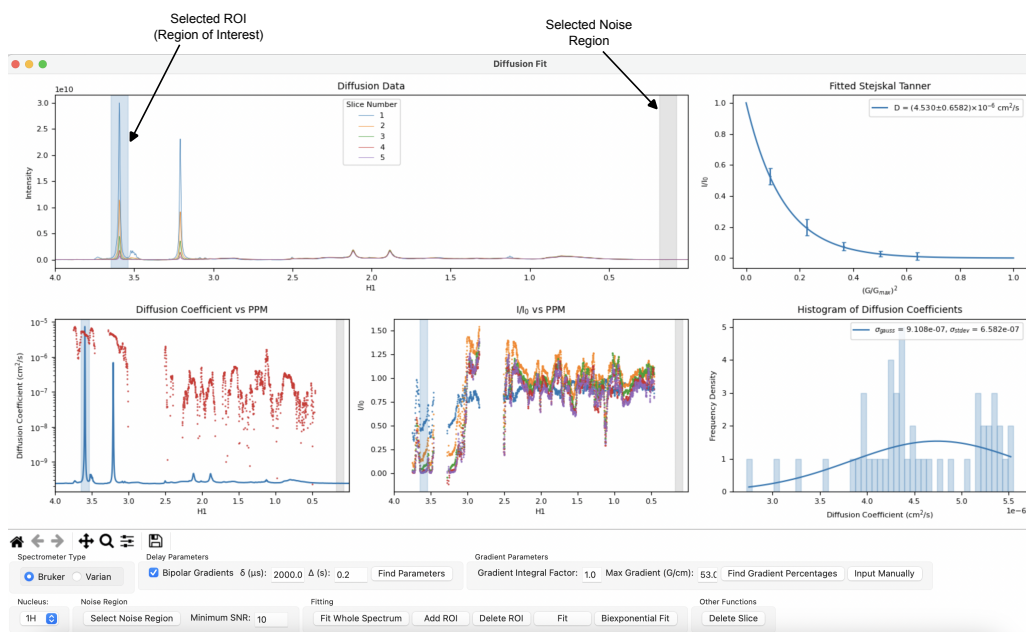


Figure 6: The diffusion frame produced when the diffusion analysis has been performed on a pseudo2D spectrum containing diffusion NMR data.

in grey. The “Fit Whole Spectrum” button can then be pressed to determine the diffusion coefficient for every point in the spectrum with an intensity greater than the Minimum SNR $\times$ Mean Intensity in the Noise Region. Regions of interest can be added using the “Add ROI” button and then dragging over a region of the spectrum. The “Fit” button can then be used to apply the Stejskal-Tanner analysis of the diffusion data in that region. Multiply ROI (regions of interest) can be added simultaneously on the same plots.

Fitting Relaxation Data

The fit relaxation window has been designed to determine longitudinal (R_1) and transverse (R_2) relaxation rates from pseudo2D relaxation data. The process of calculating relaxation rates works analogously to the diffusion frame described above.

CEST

The CEST button can be used for quick preliminary analysis of pseudo-2D CEST data (in for example a water-CEST experiment) obtained by varying the offset frequency of a saturation pulse and interleaving each on-resonance experiment with an off-resonance experiment. The CEST Z-spectrum for a given chemical shift (selected in the same way as viewing the a vertical slice of the 2D plane) can be created after pressing the “CEST analysis” button. The resulting CEST Z-spectrum is then showed which can be saved as a spectrum file to be viewed and further analysed in a 1D SpinView frame.

0.6.3 3D

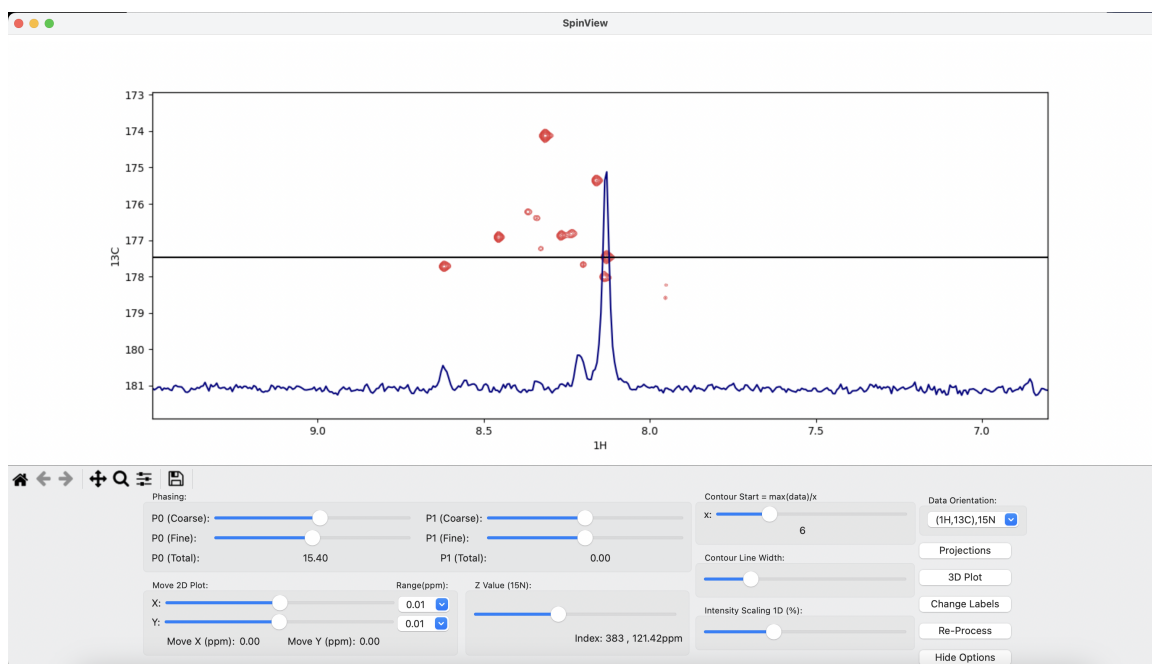


Figure 7: The SpinView window obtained when opened from a directory containing a processed 3D HNCO spectrum.

Three-dimensional NMR spectra can be opened using SpinView. Figure 7 shows the SpinView when opened on a 3D HNCO experiment. Different 2D slices of the 3D spectrum can be viewed by changing the “Z Value” slider. Furthermore, the data orientation can be altered by selecting a new orientation in the “Data Orientation” drop-down box. Many of the features in the 3D window are analogous to those in the 2D frame, including the method to obtain horizontal and vertical slices of the current contour plot using the “h” and “v” keys.

Viewing Projections

The projections button will search the current directory for nmrPipe projection (.dat) data files. These are created automatically when processing 3D data using SpinProcess. If found, it will bring out a new window containing instances of the 2D SpinView frames for the projection of each axis. Each 2D projection has the full functionality of the 2D frame, and additional spectra can be overlaid onto each projection.

Waterfall plot

If a current horizontal/vertical slice has been selected, a waterfall plot can be generated which spawns a 1D SpinView window showing an overlay of the 1D slices selected for each Z slice.

Show Bore

This module enables bores down the Z-dimension at particular locations in the current 2D planes to be viewed. From this window, the “Read Peaks” button can be pressed to open up a window to allow 3D peak picking using nmrglue. Furthermore, the peak lists can be adjusted interactively (in the x/y and z-planes in turn) in a similar manner to the two-dimension method described previously. Updated peak lists can then be saved for use in other analysis pipelines.